

AI for Economic Research: Dynamic Models, Language, and Agents

Lecture 1.1: Introduction to Quantitative Macro & Dynamic Programming

Zhigang Feng

Spring 2026

Topic 1.1 Agenda

1. **What is Quantitative Macroeconomics?**
2. **Dynamic Programming Foundations**

What is Quantitative Macroeconomics?

What is Quantitative Macroeconomics?

- A field that uses numerical methods and computational tools to solve and analyze macroeconomic models, providing precise, quantitative answers to economic questions.
- **The Methodological Triad:**
 1. *Theory*: Economic models with microfoundations
 2. *Data*: Calibration and estimation
 3. *Computation*: Numerical solution methods
- **Core Techniques:**
 - Dynamic programming and optimal control
 - Numerical optimization and root-finding
 - Monte Carlo simulation
- **This Course:** Adding machine learning to the toolkit—neural networks, stochastic optimization, and deep reinforcement learning.

What is Quantitative Macroeconomics?



Workhorse Models in Quantitative Macro

- **Representative Agent Models:**

- Neoclassical (stochastic) optimal growth model
- Real Business Cycle (RBC) model
- New Keynesian DSGE model

- **Heterogeneous Agent Models:**

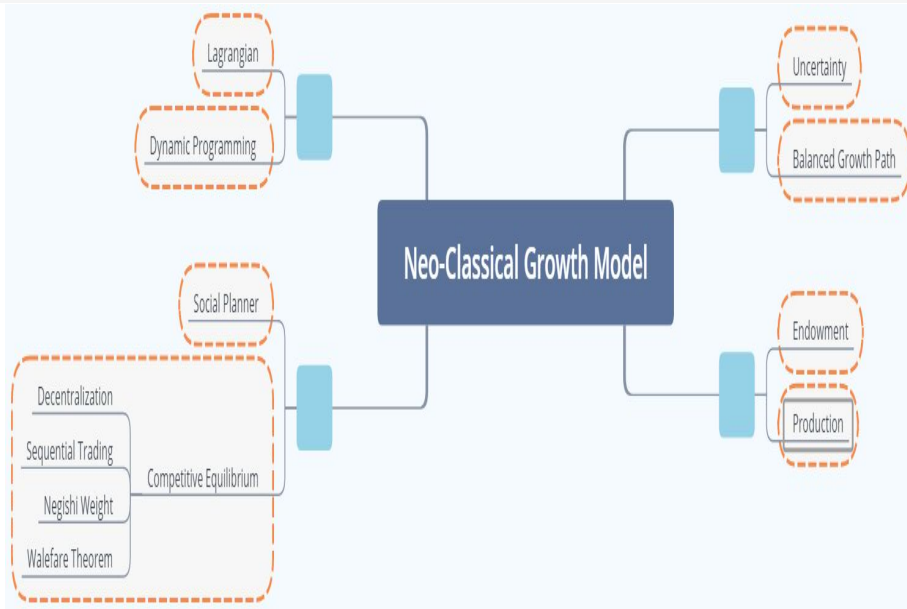
- Bewley-Aiyagari-Huggett (incomplete markets, idiosyncratic risk)
- Krusell-Smith (aggregate uncertainty + heterogeneity)
- Overlapping generations (OLG) models

- **Optimal Policy Models:**

- Ramsey taxation problems
- Time-inconsistency and commitment

- **Common Thread:** All require solving dynamic optimization problems—the Bellman equation is central.

Optimal Growth Model at the Core



Optimal Growth Model: Setup

- The household solves the following dynamic optimization problem:

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \log c_t$$

$$\text{s.t. } c_t + k_{t+1} = Ak_t^\alpha + (1 - \delta)k_t$$

$$\beta \in (0, 1), \quad A > 0, \quad \alpha \in (0, 1), \quad \delta \in [0, 1]$$

$$k_t, c_t \geq 0, \quad k_0 \text{ given}, \quad t = 0, 1, 2, \dots$$

- Recursive Formulation (Bellman Equation):**

$$v(k) = \max_{k' \in \Gamma(k)} \{\log(Ak^\alpha + (1 - \delta)k - k') + \beta v(k')\}$$

where $\Gamma(k) := \{k' \mid 0 \leq k' \leq Ak^\alpha + (1 - \delta)k\}$

- Solution:** Value function $v(k)$ and policy function $k' = g(k)$.

From Sequence Problem to Recursive Problem

- **The Challenge:** The sequence problem requires choosing an infinite sequence $\{c_t, k_{t+1}\}_{t=0}^{\infty}$ simultaneously—computationally intractable.
- **Key Insight:** The problem has *recursive structure*—optimal decisions depend only on the current state k , not on history or calendar time.
- **Bellman's Principle of Optimality (1957):**
“An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.”
- **Implication:** Replace infinite-dimensional optimization with a functional equation—the Bellman equation—reducing the problem to finding a fixed point in function space.

Dynamic Programming Foundations

The Bellman Operator

- Define the **Bellman operator** $T : \mathcal{C}(X) \rightarrow \mathcal{C}(X)$:

$$(Tv)(k) = \max_{k' \in \Gamma(k)} \{u(Ak^\alpha + (1 - \delta)k - k') + \beta v(k')\}$$

- The value function v^* is a **fixed point**: $Tv^* = v^*$

- **Central Questions:**

1. Does a fixed point exist?
 2. Is it unique?
 3. Can we compute it via iteration $v_{n+1} = Tv_n$?
 4. What properties does v^* inherit (continuity, concavity)?
- The following theoretical results answer these questions affirmatively.

Contraction Mapping Theorem

- **Definition:** Let (S, d) be a metric space. An operator $T : S \rightarrow S$ is a *contraction* with modulus $\beta \in (0, 1)$ if:

$$d(Tv, Tw) \leq \beta \cdot d(v, w) \quad \forall v, w \in S$$

- **Contraction Mapping Theorem (Banach, 1922):** If (S, d) is a complete metric space and T is a contraction, then:
 1. T has a *unique* fixed point $v^* \in S$
 2. For any $v_0 \in S$, the sequence $v_{n+1} = Tv_n$ converges to v^*
 3. Rate of convergence: $d(v_n, v^*) \leq \frac{\beta^n}{1-\beta} d(v_1, v_0)$
- **For Economics:** Guarantees that Value Function Iteration (VFI) converges to the unique solution of the Bellman equation.

Blackwell's Sufficient Conditions

- **Problem:** How do we verify that T is a contraction?
- **Blackwell (1965):** An operator $T : \mathcal{B}(X) \rightarrow \mathcal{B}(X)$ on bounded functions is a contraction if:
 1. **Monotonicity:** $v(x) \leq w(x)$ for all $x \Rightarrow (Tv)(x) \leq (Tw)(x)$
 2. **Discounting:** $\exists \beta \in (0, 1)$ such that $T(v + a)(x) \leq (Tv)(x) + \beta a$ for all $a \geq 0$
- **Verification for the Bellman Operator:**
 - Monotonicity: Higher continuation value $\Rightarrow$ higher current value
 - Discounting: The factor β in $\beta v(k')$ provides the discounting
- **Key Insight:** The discount factor $\beta < 1$ is not just economically meaningful—it is mathematically essential for convergence.

Theorem of the Maximum

- **Theorem (Berge, 1963):** (1963 English translation; original French 1959.) Let $f(x, a)$ be continuous and $\Gamma(x)$ be a continuous, compact-valued correspondence. Then:
 1. The *value function* $v^*(x) = \max_{a \in \Gamma(x)} f(x, a)$ is continuous
 2. The *policy correspondence* $G(x) = \arg \max_{a \in \Gamma(x)} f(x, a)$ is upper hemi-continuous
- **Economic Implications:**
 - Small changes in k lead to small changes in $v(k)$ and $g(k)$
 - Ensures stability of computed solutions
 - Justifies numerical approximation on discrete grids
- **Additional Properties:** Under strict concavity of $u(\cdot)$ and a convex constraint graph (i.e. Γ has convex graph, so the feasible set is convex):
 - Policy correspondence is single-valued: a *function* $g(k)$
 - Value function $v(k)$ is strictly concave

Theoretical Foundation: Summary

Result	What It Guarantees
Principle of Optimality	Recursive formulation is valid
Contraction Mapping Thm	Existence, uniqueness, convergence
Blackwell's Conditions	Easy verification of contraction
Theorem of the Maximum	Continuity and stability

Together, these results establish:

1. The Bellman equation has a *unique* solution v^*
2. We can *compute* v^* by iterating $v_{n+1} = Tv_n$
3. The solution is *well-behaved* (continuous, concave)
4. The policy function $g(k)$ is *stable* under perturbations

⇒ **Solid foundation for computational methods**